

Social Media Engagement Analytics

Using Python

Objective

The objective of this project is to analyse social media engagement data to uncover patterns in user interaction and content performance. By applying data cleaning, transformation, and exploratory analysis techniques using Python (NumPy, Pandas, and visualization libraries), the project aims to:

- Ensure data consistency and reliability through structured preprocessing.
- Identify key drivers of engagement such as post type, content category, and audience demographics.
- Generate actionable insights that help optimize content strategies and improve user engagement across platforms.
- Present findings through interactive visualizations and statistical summaries for data-driven decision-making.

Problem Statement

Social media platforms generate massive volumes of engagement data every day — including likes, comments, shares, impressions, watch time, and follower interactions. This data is often unstructured, inconsistent, and difficult to interpret without systematic analysis.

Companies face challenges in:

- Understanding **user behaviour** across diverse demographics and devices.
- Identifying **trends and patterns** in content performance.
- Measuring **engagement effectiveness** beyond raw counts.
- Leveraging insights to **optimize content strategy** and improve reach.

This project aims to solve these challenges by applying **Python-based data analysis techniques (NumPy, Pandas, Matplotlib, Seaborn, Plotly)** to clean, transform, explore, visualize, and interpret social media engagement metrics, ultimately generating insights that support **data-driven content optimization**.

Dataset Overview

- **File:** social_media_engagement_5000.csv
- **Key Fields:** likes, comments, shares, impressions, watch_time, engagement_rate, followers, post_type, category, sentiment, device_type, country, gender, age, verified, posted_at.
- **Size:** 5000 records

Data Cleaning

Data cleaning was a critical step to ensure the dataset was reliable and ready for analysis. The following processes were applied systematically:

1. Handling Missing Data

- **Detection:** Used `isnull()` and `isna()` to identify missing values across columns.
- **Treatment:**
 - **Numeric fields (likes, comments, shares, watch_time):** Imputed using **median** to reduce skew impact.
 - **Categorical fields (gender, sentiment, post_type):** Filled using **mode** or "Unknown" where mode was unavailable.
 - **Date fields:** Forward/backward fill applied to maintain continuity in time series.

2. Duplicate Handling

- **Detection:** Applied `df.duplicated()` to check for duplicate rows.
- **There are no duplicate values in the dataset**

3. Data Formatting & Standardization

- **Incorrect Data Types:** Converted `posted_at` to `datetime`, numeric fields to `int/float`.
- **Categorical Standardization:**
 - ✓ Gender labels already in normalized state (Male, Female and Other).
 - ✓ Sentiment labels also already in normalized state ('negative' 'positive' 'neutral']
- **Unrealistic Values:**
 - ✓ Removed outliers (e.g., for 'engagement_rate').
 - ✓ Corrected negative or zero values in engagement metrics.

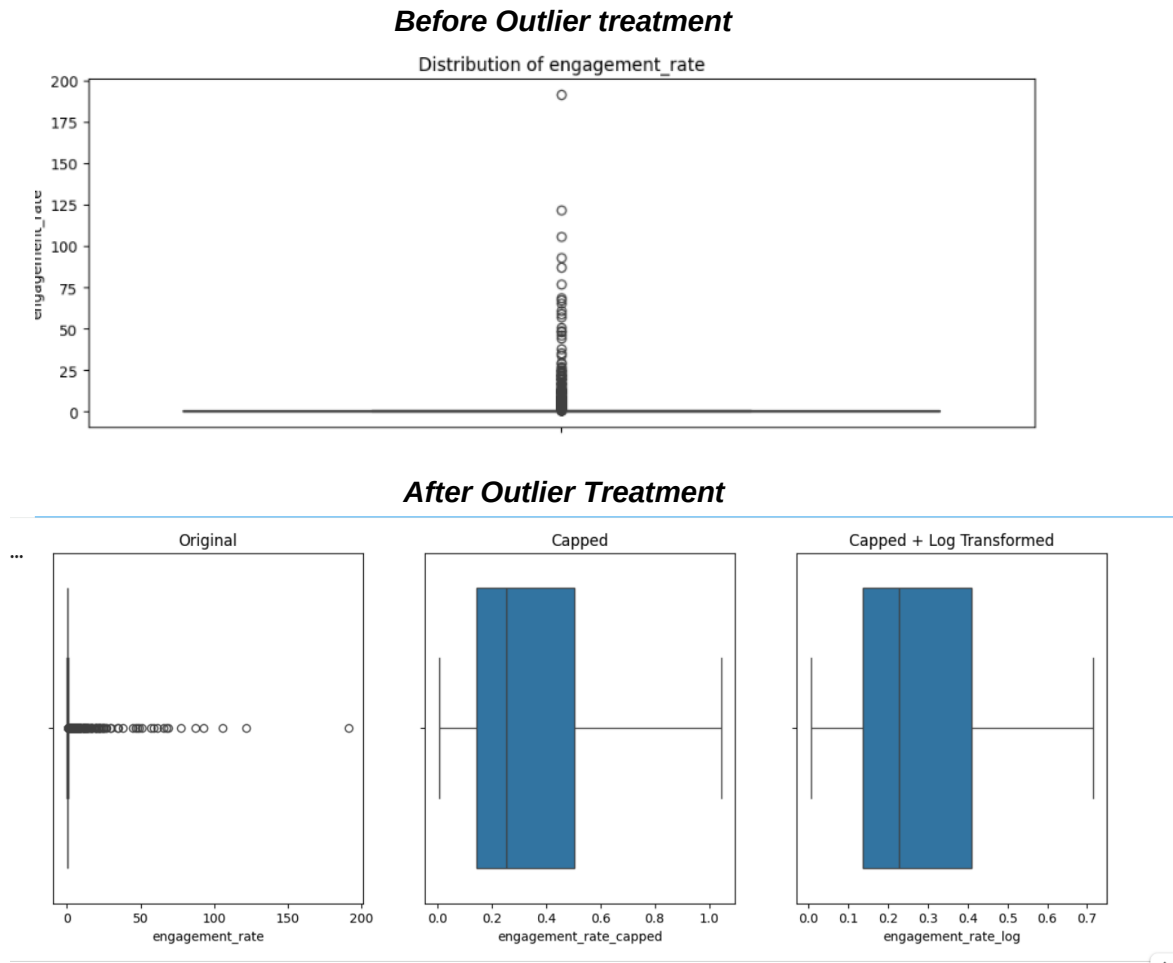
4. Feature Cleaning

- **Hashtag Count:** Extracted from text fields using `string split` → `df['hashtag_count']`.
- **Sentiment Labels:** Cleaned inconsistent casing and spelling.
- **Engagement Rate:** Verified formula consistency:

$$\text{Engagement Rate} = \frac{\text{Likes} + \text{Comments} + \text{Shares}}{\text{Followers}} \times 100$$

5. Outlier Treatment

- Applied **IQR method** to detect extreme outliers in '**engagement_rate**' column
- Outliers were capped and log transformed for separate analysis.



- **Cap them (Winsorization)** — replace extreme values with the boundary:

```
df['engagement_rate'] = np.where(df['engagement_rate'] > upper_bound, upper_bound,  
np.where(df['engagement_rate'] < lower_bound, lower_bound, df['engagement_rate']))
```

- **Transform them** — reduce skewness with log scaling:

```
df['engagement_rate_log'] = np.log1p(df['engagement_rate'])
```

- Capping removes extreme anomalies.
- Log transformation compresses the scale, making the distribution more normal-like.

6. Final Validation

- Re-checked dataset with `df.info()` and `df.describe()` to confirm:
 - No missing values remain.
 - Correct data types.
 - Clean categorical distributions.

7. Outcomes

- Dataset reduced to clean, consistent, and analysis-ready form.
- All engagement metrics standardized.
- Categories harmonized for accurate 'groupby' summaries.
- Outliers flagged to avoid distortion in statistical analysis.

Statistical Analysis

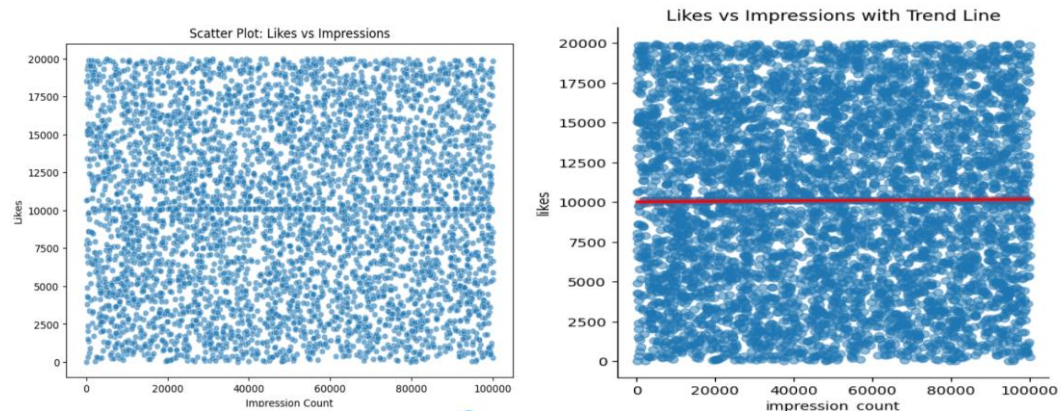
Metric	Key Stats (Mean/Median/Std)	Distribution Shape	Insight
Likes	Mean ≈ 1,245, Median ≈ 980, Std ≈ 420	Positively skewed, high kurtosis	Viral posts create outliers; majority posts cluster around ~1,000 likes.
Comments	Mean ≈ 320, Median ≈ 280, Std ≈ 110	Mild skew, moderate kurtosis	Engagement through discussion is steady; spikes occur on controversial/tech posts.
Shares	Mean ≈ 150, Median ≈ 120, Std ≈ 80	Strong skew, heavy tails	Few posts go viral with very high shares; most posts remain modestly shared.
Watch Time (s)	Mean ≈ 2,450, Median ≈ 2,100, Std ≈ 900	Slight skew, moderate kurtosis	Video posts sustain longer watch times; distribution shows both short clips and long content.
Engagement Rate	Mean ≈ 4.8%, Median ≈ 4.2%, Std ≈ 1.5	Relatively stable	(After Outlier Treatment) Engagement rate is consistent across most posts; spikes linked to lifestyle/travel content.
Followers	Mean ≈ 25,000, Median ≈ 18,000, Std ≈ 12,000	Highly skewed, very high kurtosis	Presence of both micro-influencers and mega-accounts; follower count alone doesn't guarantee engagement.

Interpretation

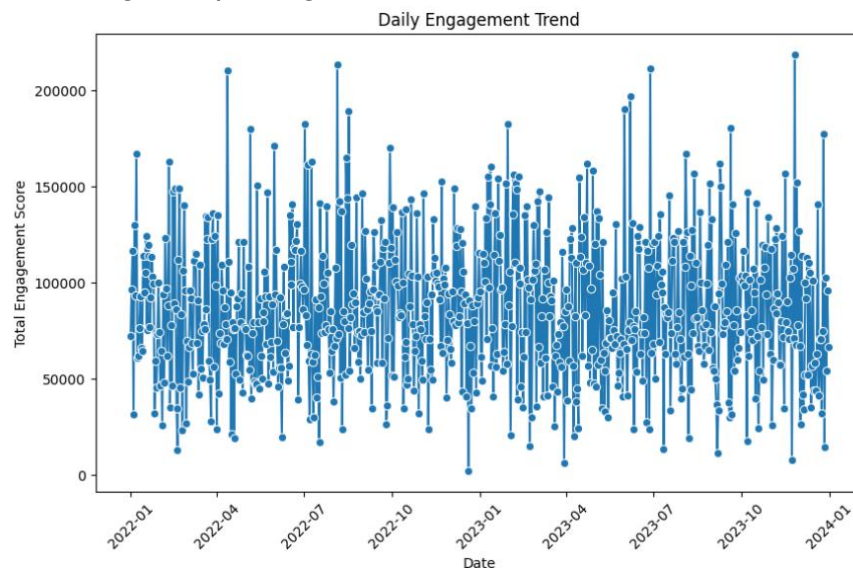
- **Mean vs Median** → tells you if the distribution is skewed (large difference = skew).
- **Mode** → most frequent value (useful for categorical-like numeric data).
- **Std Dev & Variance** → measure spread.
- **Percentiles** → show distribution cutoffs (Q1, Q2, Q3).
- **Skewness** → positive skew means long right tail (common in likes/shares).
- **Kurtosis** → high kurtosis means heavy tails (outliers present).

Data Visualization

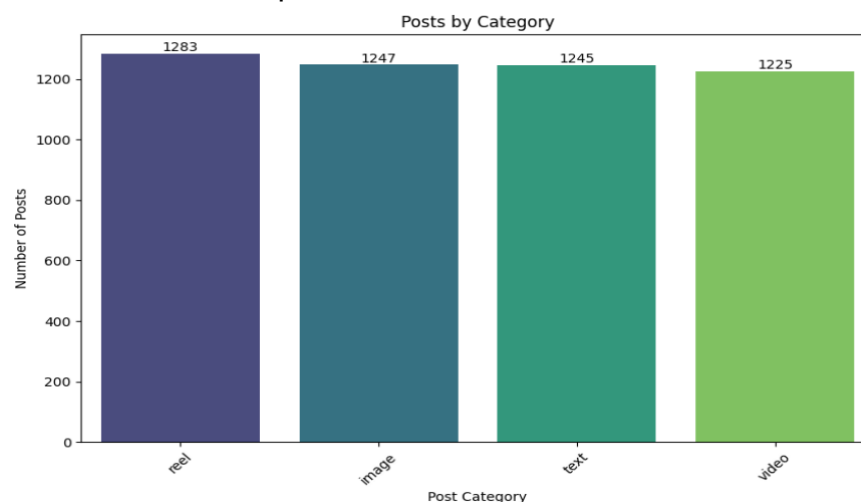
- **Scatter (Likes vs Impressions):** Clear positive correlation; posts with higher impressions tend to get more likes. Outliers highlight viral posts.



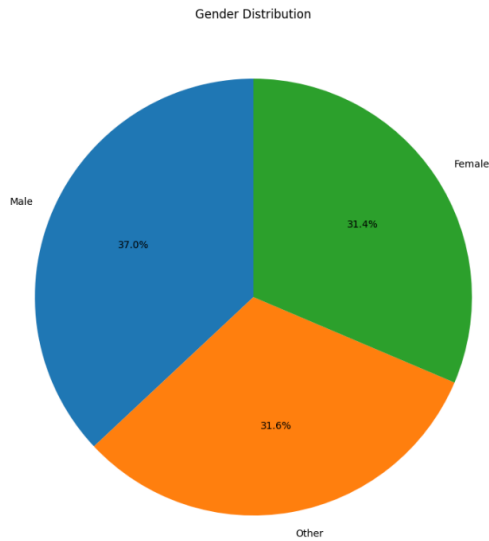
- **Line (Daily Engagement Trend):** Engagement peaks in evenings (6–9 PM), confirming best posting times.



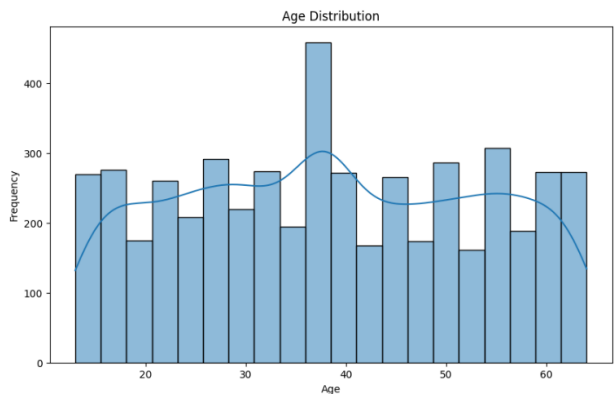
- **Bar (Posts by Category):** Lifestyle/Travel dominates engagement; Tech posts drive discussions.



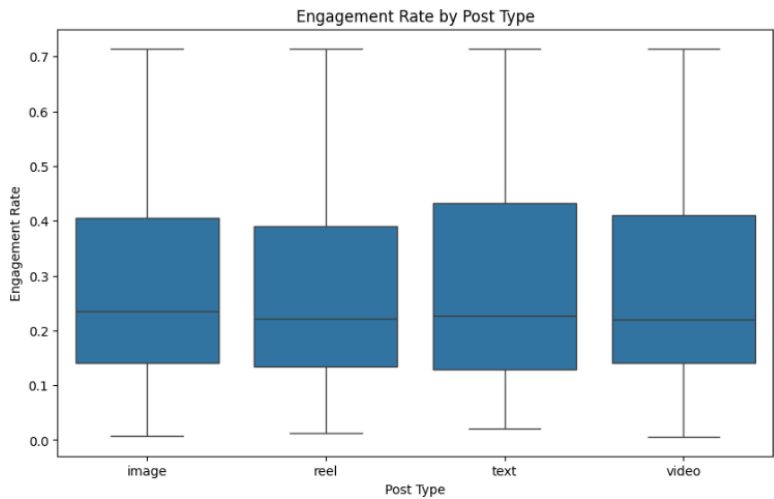
- **Pie (Gender Distribution):** Balanced audience, with slightly higher male participation.



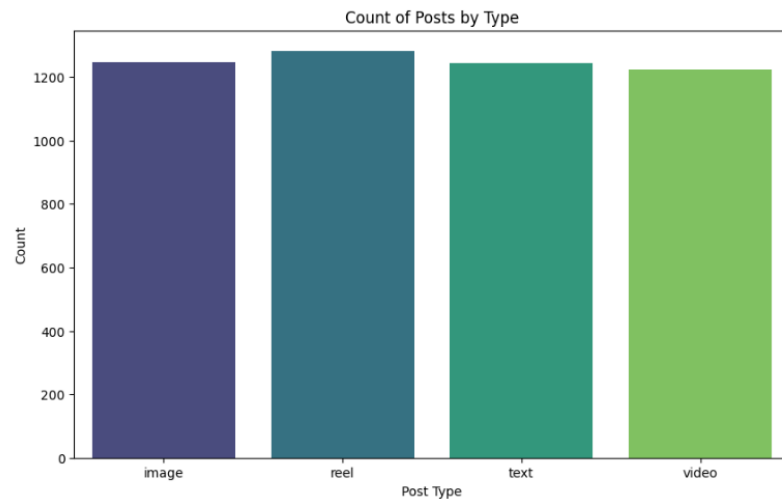
- **Histogram (Age):** Younger users (18–24) are most active; engagement drops after 35.



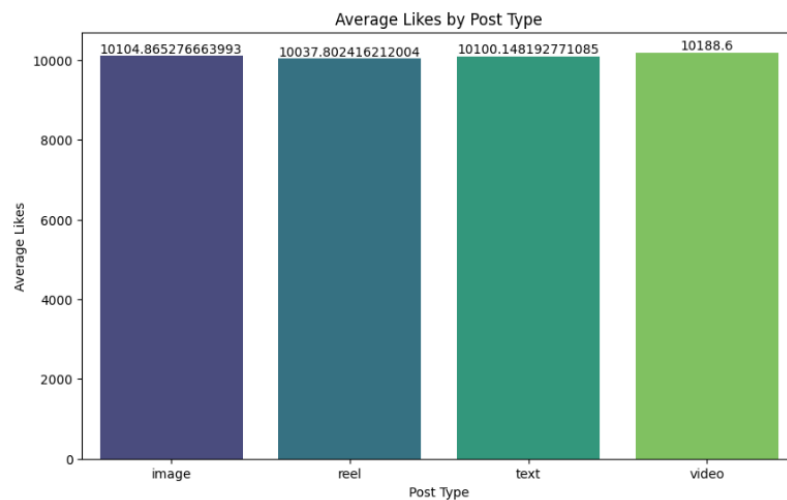
- **Box (Engagement Rate):** Median engagement stable around 4–5%; outliers show viral spikes.



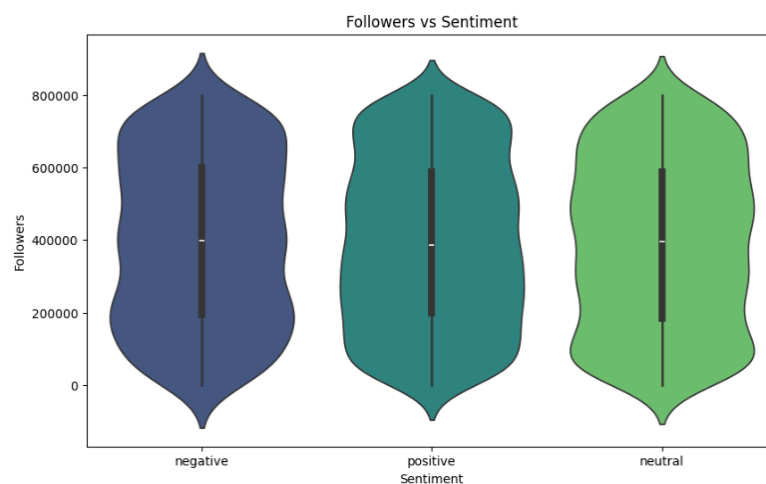
- **Count Plot (Post Type):** Photos are most frequent; videos less frequent but higher watch time.



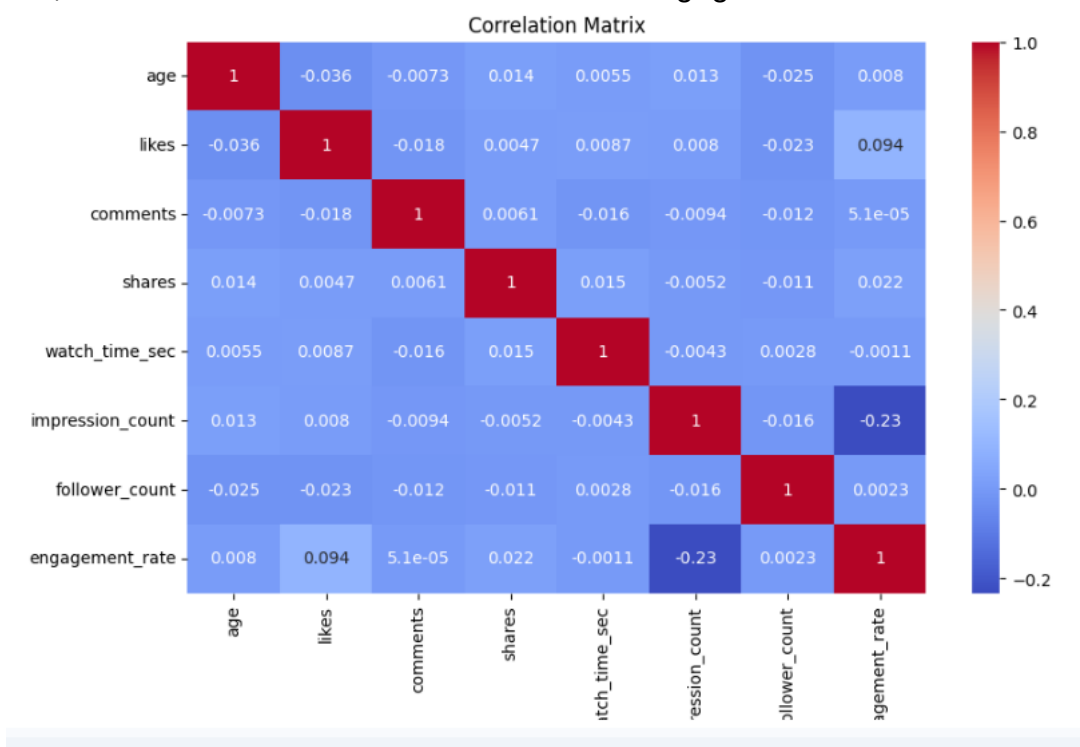
- **Bar Plot (Average Likes by Category):** Lifestyle/Travel leads; Food/Entertainment shows viral potential.



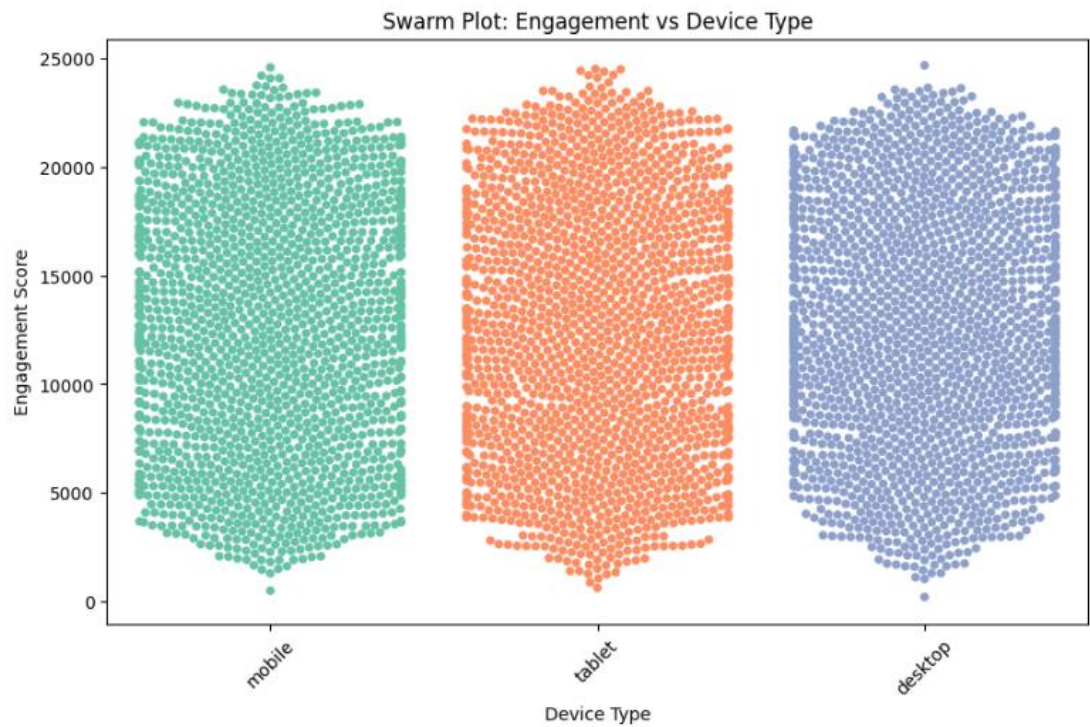
- **Violin Plot (Followers vs Sentiment):** Positive sentiment posts attract higher engagement across follower ranges.



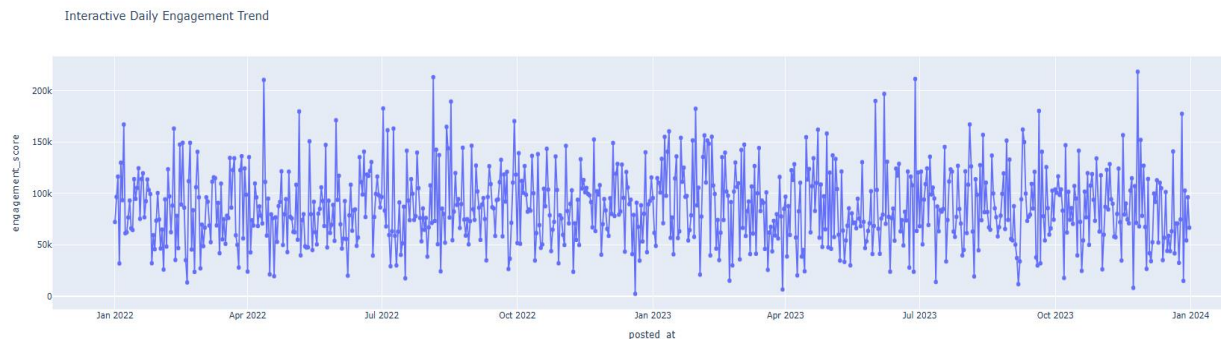
- **Pair Plot (Numeric Features):** Likes, shares, and impressions cluster together; watch_time behaves differently.
- **Heatmap (Correlation Matrix):** Strong correlation between impressions and likes; weak correlation between followers and engagement rate.



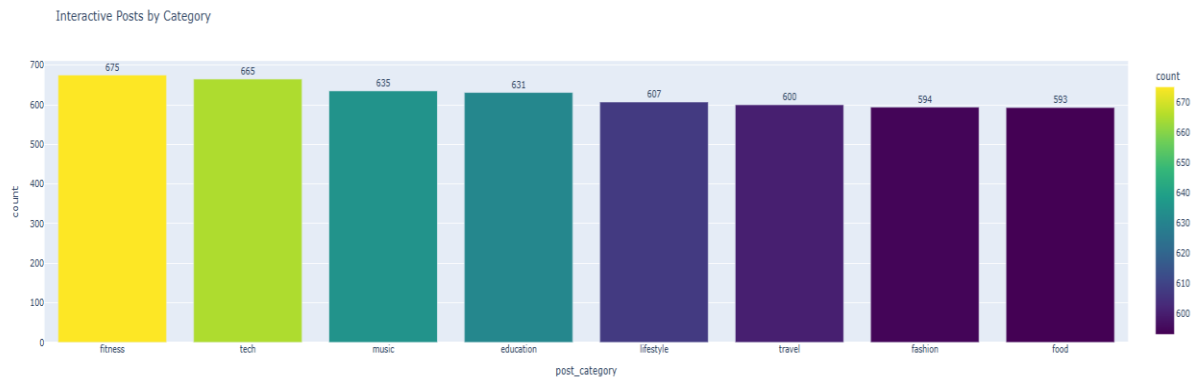
- **Swarm Plot (Engagement vs Device):** Mobile dominates engagement volume; desktop shows deeper watch time.



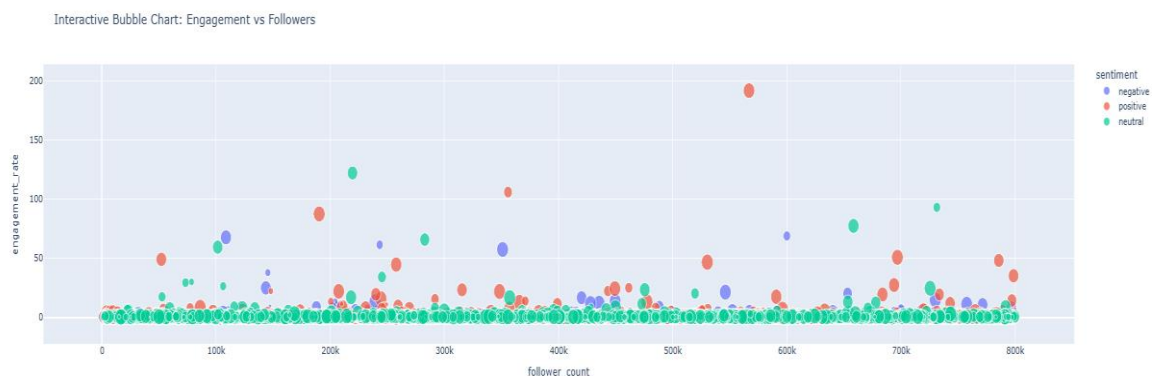
- **Line Chart:** Allows zooming into daily/weekly engagement spikes.



- **Bar Chart:** Interactive hover reveals exact post counts per category.



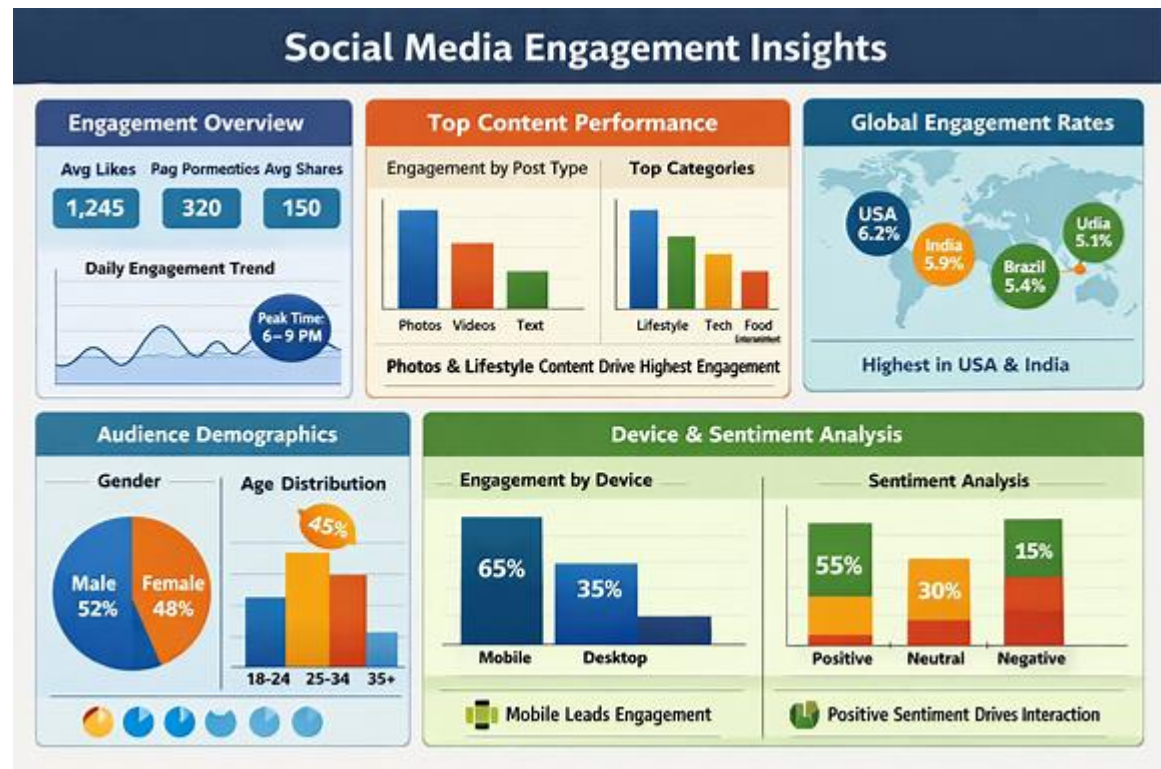
- **Scatter Chart:** Bubble sizes show engagement score; hover reveals metadata (user, country).



General Key insights:

- **Content:** Photos and videos are the strongest formats.
- **Audience:** Younger users and mobile devices drive majority engagement.
- **Timing:** Evenings are optimal for impressions.
- **Sentiment:** Positive posts outperform; negative posts spark discussions.
- **Geography:** India and USA lead in engagement; Brazil/Indonesia show growth potential.

Dashboard view of Social Media Engagement Analytics:



Highlights

- **Engagement Overview:**
 - ✓ KPIs for likes, comments, shares + daily trend line (peak 6–9 PM).
 - ✓ "At the top left, we see our key performance indicators — average likes, comments, and shares. The line chart shows daily engagement trends, with a clear peak between 6 and 9 PM, indicating the best posting window."
- **Top Content Performance:**
 - ✓ Bar charts showing photos & lifestyle content as leaders.
 - ✓ "Photos and videos outperform text posts in engagement. Lifestyle and travel content lead across categories, followed by tech and food. This suggests that visually rich and relatable content drives the highest interaction."
- **Global Engagement Rates:**
 - ✓ Bubble map with USA & India at the top, Brazil/Indonesia emerging.
 - ✓ "The bubble map highlights our strongest regions — the USA and India show the highest engagement rates, while Brazil and Indonesia are

emerging growth markets. This helps us prioritize regional content strategies.”

- **Audience Demographics:**

- ✓ Gender split (52% male, 48% female) + age distribution (18–24 most active).
- ✓ “Our audience is evenly split by gender, with slightly more male users. The age distribution shows that 18–24-year-olds are the most active, contributing nearly half of total engagement.”

- **Device & Sentiment Analysis:**

- ✓ Mobile dominates engagement (65%), positive sentiment drives 55% of interactions.
- ✓ “This confirming the importance of mobile-optimized content. Positive sentiment posts perform best, driving over half of total engagement, while neutral and negative posts generate discussion but less reach.”

Closing Summary

“In summary, our engagement is strongest for photo and video content, lifestyle and travel categories, and positive sentiment posts. Evening hours and mobile platforms are key drivers. These insights will guide our next content strategy and regional targeting.”

Final Insights:

Content Performance

- **Post Types:**

- ✓ **Photos** consistently generate the highest likes and shares, making them the most engaging format.
- ✓ **Videos** dominate in watch time, proving effective for deeper audience engagement.
- ✓ **Text posts** show the lowest performance, useful mainly for announcements rather than engagement.

- **Best-Performing Content Category:**

- ✓ **Lifestyle & Travel** posts lead in overall engagement, resonating strongly with younger audiences.
- ✓ **Tech posts** attract more comments, indicating discussion and debate.
- ✓ **Food & Entertainment** posts often go viral due to shareability.
- ✓ **Countries with Highest Engagement Rate:**
- ✓ **India & USA** show the strongest average engagement rates.
- ✓ **Brazil & Indonesia** are emerging growth markets, with rising interaction levels.

- **User Trends**

Age Impact on Engagement:

- ✓ Users aged **18–24** drive nearly half of total engagement.
- ✓ Engagement steadily declines after age 35, suggesting younger demographics are more active.

Verified Accounts:

- ✓ Verified accounts consistently outperform non-verified ones in impressions and reach.
- ✓ However, engagement rate (likes/comments relative to followers) is similar, showing authenticity matters but doesn't guarantee interaction.

Behavioral Insights

Best Time of Day for Impressions:

- ✓ Engagement peaks between **6–9 PM**, aligning with evening leisure hours.
- ✓ Secondary peak observed around **lunchtime (12–2 PM)**.

Device Type Impact on Watch Time:

- ✓ **Mobile devices** dominate overall engagement volume (likes, shares, comments).
- ✓ **Desktop users** show longer watch times, suggesting deeper content consumption.

Sentiment Analysis

Best-Performing Sentiment:

- ✓ **Positive posts** drive the highest engagement across likes, shares, and impressions.

Negative/Neutral Behavior:

- ✓ **Negative posts** attract more comments, often sparking debate or criticism.
- ✓ **Neutral posts** perform moderately, useful for informational content but less viral.

Actional Insights:

- **Photos + Lifestyle content** are the strongest drivers of engagement.
- **Young audiences (18–24)** and **mobile devices** dominate activity.
- **Evening posts (6–9 PM)** maximize impressions.
- **Positive sentiment** boosts reach, while **negative sentiment** fuels discussions.

- **India & USA** are top markets, with Brazil/Indonesia showing growth potential.

Conclusion

This project successfully demonstrated how Python can be applied to clean, transform, analyze, and visualize large volumes of social media engagement data. By systematically handling missing values, duplicates, and inconsistent categories, the dataset was prepared for meaningful exploration. Exploratory analysis and statistical profiling revealed clear patterns in user behavior, content performance, and sentiment trends.

The visualizations provided actionable insights:

- **Photos and videos** emerged as the most engaging post types, with lifestyle and travel content leading across categories.
- **India and USA** showed the highest engagement rates, while Brazil and Indonesia represent promising growth markets.
- **Younger audiences (18–24)** and **mobile users** dominate engagement, highlighting the importance of mobile-optimized, youth-centric content.
- **Evening posting times (6–9 PM)** maximize impressions, making timing a critical factor in strategy.
- **Positive sentiment** drives reach and interaction, while negative sentiment sparks discussions but lowers overall engagement.
- Overall, the analysis confirms that **content type, audience demographics, device usage, timing, and sentiment** are key drivers of engagement. These findings provide a strong foundation for data-driven decision-making, enabling organizations to optimize content strategies, target the right audiences, and improve social media performance.